

Module/Pathway/Taxon Review: Bacterial Chromosomal DNA Replication in *Pseudomonas putida* KT2440 (KEGG ppu03030)

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu03030 "DNA replication" (local module: bacterial chromosome-replication) **Purpose:** Support manual module satisfiability and gene-annotation curation.

1. Executive Summary

The KEGG ppu03030 "DNA replication" module is **fully satisfiable (COVERED)** in *Pseudomonas putida* KT2440. All seven expected steps of the reusable bacterial chromosome-replication module — (1) DnaA-dependent origin initiation, (2) DnaB/SSB fork unwinding and single-strand protection, (3) DnaG RNA priming, (4) Pol III α/ϵ processive synthesis and proofreading, (5) DnaN β -clamp processivity, (6) DnaX clamp-loader assembly with $\delta/\delta'/\chi/\psi$, and (7) LigA Okazaki-fragment ligation — have high-confidence, one-to-one gene assignments in the KT2440 genome. The core replisome is completely encoded and internally consistent in UniProt annotation.

The single most important curation action is that the **replication initiator** dnaA (PP_0010, UniProt P0A116) is a **metadata gap, not a biological gap**. DnaA is present in the KT2440 genome — it sits immediately upstream of dnaN/PP_0011 in the canonical *dnaA-dnaN-recF-gyrB* origin-proximal cluster — but it is omitted from the 18-gene candidate list because the KEGG map03030 reference pathway does not enumerate the initiator. This locus should be added to the module metadata as a covered step.

The second key finding is that **three candidate genes are likely over-propagated / paralogous annotations that do not belong to the core replisome**: PP_0353 (Q88QY1), PP_4768 (Q88DQ5), and PP_3893 (Q88G33). Domain-signature analysis (InterPro/Pfam) cleanly discriminates these from the authentic components: PP_0353 and PP_4768 carry only the

generic RNase H-like/DEDDh exonuclease fold and lack the DnaQ-proteobacterial epsilon signature (IPR006309) that uniquely marks the true proofreader `dnaQ`/PP_4141; PP_3893 carries a phage-type DnaB helicase signature (IPR019889) absent from the bona fide replicative helicase `dnaB`/PP_4873. These three should be demoted to **candidate_uncertain** and promoted to full `fetch-gene` review, but their removal does not create any module gap because every step they nominally covered is already satisfied by a higher-confidence gene. Finally, enterobacterial-lineage-specific factors (`dnaC`/`dnaI`, `priB`/`priC`/`dnaT`, `seqA`) are legitimately **not_expected_in_target_taxon**.

2. Target-Organism Pathway Definition

Process included. KEGG `ppu03030` as scoped for this module covers *chromosomal* (replisome-centric) DNA replication: origin initiation, replication-fork unwinding and single-strand protection, RNA primer synthesis, processive leading/lagging-strand DNA synthesis with proofreading, sliding-clamp processivity, clamp loading and replisome coupling, and Okazaki-fragment maturation/nick sealing.

Neighboring pathways to keep separate. KT2440's candidate metadata already reveals the boundary tension: several genes carry a **primary bucket of kegg:ppu03420 (Nucleotide excision repair)** rather than `ppu03030` — notably `polA`/PP_0123 (DNA polymerase I), `ligA`/PP_4274 (NAD-dependent DNA ligase), and `ligB`/PP_4968 (DNA ligase B). These are shared "primer-processing and nick-sealing" enzymes that participate in both replication (Okazaki maturation) and repair. For a chromosomal-replication module, `polA`, `ligA`, `rnhA`, and `rnhB` are best treated as **supporting** primer-processing functions rather than core replisome subunits. Broad overview maps (`map01100` metabolic pathways; recombination/repair maps `ppu03430`/`ppu03440`) should be kept out of scope.

Alternate names / database definitions. The pathway is variously called "DNA replication" (KEGG `map03030`), the "bacterial replisome," or "DNA polymerase III holoenzyme + accessory factors." The initiator `DnaA` belongs to KEGG's replication-*initiation* description but is not enumerated in the `map03030` enzyme list, which is why it is absent from the candidate metadata despite being biologically central. Curators should note this systematic KEGG-map limitation.

3. Expected Step Model

The reusable module and its ordering map onto KT2440 loci as follows:

#	Module step	Expected activity	KT2440 gene	Locus	UniProt	Verdict
1	Origin initiation	DNA replication origin binding	dnaA	PP_0010	P0A116	Covered (add to metadata)
2	Fork unwinding	5'→3' replicative helicase	dnaB	PP_4873	Q88DF2	Covered
2	Single-strand protection	ssDNA binding	ssb	PP_0485	Q88QK5	Covered
3	RNA priming	DNA-directed RNA primer synthesis	dnaG	PP_0388	P0A118	Covered
4	Processive synthesis	DNA-directed DNA polymerase (Pol III α)	dnaEA	PP_1606	Q88MG5	Covered
4	Proofreading	3'→5' exonuclease (Pol III ϵ)	dnaQ	PP_4141	Q88FF6	Covered
5	Sliding-clamp processivity	β -clamp processivity factor	dnaN	PP_0011	P0A120	Covered
6	Clamp loading	DnaX τ/γ clamp-loader ATPase	dnaX	PP_4269	Q88F30	Covered
6	Clamp loading	δ subunit	holA	PP_4796	Q88DM9	Covered
6		δ' subunit	holB	PP_1966	Q88LG7	Covered

#	Module step	Expected activity	KT2440 gene	Locus	UniProt	Verdict
	Clamp loading					
6	Clamp loading	χ subunit	holC	PP_0978/0979	Q88P75/ Q88P74	Covered (annotation swap)
6	Clamp loading	ψ subunit	holD	PP_0978/0979	(adjacent)	Candidate_uncertain
6	Core coupling	θ subunit	holE	—	—	Candidate_uncertain / not required
7	Nick sealing	NAD-dependent DNA ligase	ligA	PP_4274	Q88F25	Covered (supporting)
S	Primer processing	Pol I (nick translation)	polA	PP_0123	Q88RK6	Supporting
S	Primer removal	RNase HI / HII	rnhA / rnhB	PP_4142 / PP_1605	Q88FF5 / Q88MG6	Supporting

4. Candidate Genes and Evidence

4.1 Origin initiation — the DnaA metadata gap (Finding F001)

dnaA (PP_0010, UniProt P0A116) is a reviewed UniProt entry, "Chromosomal replication initiator protein DnaA," in proteome UP000000556 (taxon 160488). It is **absent from the 18-gene candidate list** even though it is the entry point of the entire pathway. The evidence that this is a metadata artifact rather than a biological absence is strong and multi-pronged: (i) DnaA is a reviewed, high-confidence entry in the target proteome; (ii) it occupies the canonical origin-proximal position immediately upstream of **dnaN**/PP_0011 (which is in the candidate list), following the conserved *dnaA–dnaN–recF–gyrB* synteny (*gyrB* = PP_0013, confirmed present); and (iii) the omission traces directly to KEGG map03030 not enumerating the initiator. **Curation action: add PP_0010 to module metadata as a covered Step-1 gene.**

Supporting context from KT2440 chromosome biology: origin-proximal coordination of replication and segregation is an established feature of this organism, consistent with a functional DnaA-dependent origin ([PMID: 30352930](#)).

4.2 Core replisome, steps 2–7 (Finding F002)

The core replisome is **fully encoded and consistently annotated** in UniProt. Each subunit has a specific, non-generic functional call:

- **dnaB**/PP_4873 — "main replicative DNA helicase," DnaB family, 5'→3' unwinding (Step 2).
- **ssb**/PP_0485 — single-stranded DNA-binding protein (Step 2).
- **dnaG**/PP_0388 — DnaG primase, EC 2.7.7.101 (Step 3).
- **dnaEA**/PP_1606 — Pol III α subunit, EC 2.7.7.7 (Step 4, synthesis).
- **dnaQ**/PP_4141 — Pol III ϵ subunit, 3'→5' proofreading exonuclease (Step 4, proofreading).
- **dnaN**/PP_0011 — β sliding clamp, processivity factor (Step 5).
- **dnaX**/PP_4269 — τ/γ clamp-loader ATPase, DnaX family (Step 6).
- **holA**/PP_4796 (δ), **holB**/PP_1966 (δ'), **holC** (χ) — clamp-loader accessory subunits (Step 6).
- **ligA**/PP_4274 — NAD-dependent DNA ligase LigA (Step 7).
- **Supporting:** **polA**/PP_0123 (Pol I), **rnhA**/PP_4142 (RNase HI), **rnhB**/PP_1605 (RNase HII) for Okazaki primer processing.

These are all high-confidence, one-to-one assignments and require no promotion to deep review except where noted below.

4.3 The χ/ψ (holC/holD) annotation swap (Finding F005)

The clamp-loader χ and ψ subunits map to the adjacent locus pair PP_0978–PP_0979, but with a curation-relevant inconsistency:

- **Q88P75 / PP_0978** = protein name "DNA polymerase III subunit chi" (121 aa) but carries **no gene name**.
- **Q88P74 / PP_0979** = gene name **holC** but a **blank protein name** (142 aa).

The candidate metadata assigns "chi = holC" to PP_0979, which conflicts with UniProt's chi call on PP_0978. Both loci fall in the size range of χ (~147 aa) and ψ (~137 aa) and are adjacent — the canonical *holC*–*holD* arrangement. A gene-name search for **holD**/ ψ returns nothing, but the PP_0978–PP_0979 pair is the expected χ – ψ cassette. **Interpretation:** χ is present (locus

identity needs harmonizing between UniProt and metadata), and ψ (hold) is very likely present as the unnamed partner locus. Curation action: disambiguate the PP_0978/PP_0979 pair and assign χ and ψ explicitly. This is why hold is marked **candidate_uncertain** in the step model — present by synteny/size but not name-confirmed.

4.4 Three over-propagated candidates (Findings F003, F007)

Three candidate genes carry generic or paralogous annotations and are **not** core replisome components:

PP_0353 (Q88QY1) — UniProt has no protein name, no EC number, no family assignment, and protein-existence level 4 (Predicted): a bare "Exonuclease" call. Domain analysis shows it shares only the generic RNase H-like/DEDDh exonuclease fold (Pfam PF00929; InterPro IPR013520/IPR012337/IPR036397) and **lacks the DnaQ-proteobacterial epsilon signature IPR006309**. It is an ExoX/oligoribonuclease-type exonuclease, not a replisome subunit.

PP_4768 (Q88DQ5) — carries only generic DnaQ/Pol-III-epsilon boilerplate function text but is **not** the epsilon subunit (the true ϵ is `dnaQ/PP_4141`). It is 203 aa, shares the generic RNase H-like/DEDDh fold, and — critically — **lacks IPR006309**. This is a DnaQ-family exonuclease paralog (ExoX-like), a classic case of over-propagated epsilon annotation.

PP_3893 (Q88G33) — annotated "DNA 5'→3' helicase," DnaB subfamily. It shares the DnaB Pfam architecture (PF00772 + PF03796) with the authentic `dnaB/PP_4873` but **additionally carries the phage-specific signature IPR019889 (DNA_helicase_DnaB-like_phg)** that `dnaB/PP_4873` lacks. This marks PP_3893 as a phage/prophage-type DnaB paralog, not the chromosomal replicative helicase.

The discriminating logic is summarized below:

Authentic proofreader	dnaQ/PP_4141 (252 aa): DEDDh fold + IPR006054 + IPR006309	[OK] epsilon
Over-annotation	PP_4768 (203 aa): DEDDh fold, NO IPR006309	[X] ExoX-like
Over-annotation	PP_0353 (236 aa): DEDDh fold, NO name/EC/family, PE4	[X] predicted
Authentic helicase	dnaB/PP_4873: PF00772+PF03796, NO phage signature	[OK] replicat
Over-annotation	PP_3893 (465): PF00772+PF03796 + IPR019889 (phage)	[X] phage-ty

Curation action: demote PP_0353, PP_4768, PP_3893 to **candidate_uncertain**, remove from the core replisome, and promote all three to full `fetch-gene` review. Because Steps 4 (proofreading) and 2 (helicase) are already satisfied by `dnaQ/PP_4141` and `dnaB/PP_4873`, their removal creates **no module gap**.

4.5 Lineage-specific replication features (Finding F004)

A gene-name survey of taxon 160488 in UniProt shows KT2440's replication accessory landscape differs from the enterobacterial paradigm:

Present (beyond the candidate list): `dnaE2`/PP_3119 (error-prone/translesion DNA polymerase) and `imuB`/PP_3118 (part of the *imuA-imuB-dnaE2* mutagenesis cassette); `hda`/PP_1668 (DnaA regulatory inactivation, RIDA); `priA`/PP_5088 (replication-fork restart); topoisomerases `gyrA`/PP_1767, `gyrB`/PP_0013, `parC`/PP_4912, `parE`/PP_4915; helicases/nucleases `rep`/PP_5264, `recD`/PP_4672, `recQ`/PP_4516.

Absent by name (enterobacterial-specific, not_expected_in_target_taxon): `dnaC` (and `dnaI`), `priB`, `priC`, `dnaT`, `holE`, `imuA`, `imuC`, `seqA`. The absence of `dnaC`, the *priB/priC/dnaT* primosome accessories, and `seqA` reflects genuine lineage divergence — *Pseudomonas* uses a DnaC-independent helicase-loading mechanism and lacks the *E. coli* SeqA sequestration system. These should not be counted as module gaps.

5. Gaps, Ambiguities, and Likely Over-Annotations

Item	Category	Rationale
dnaA / PP_0010	Metadata gap (add)	Present in genome; omitted only because KEGG map03030 doesn't list the initiator
holC/hold (PP_0978– PP_0979)	Ambiguity (harmonize)	χ named on wrong locus; ψ present but unnamed; needs disambiguation
holE (θ)	candidate_uncertain / not required	No name hit; θ is dispensable and often absent outside enterobacteria
PP_0353	Over-annotation → candidate_uncertain	Predicted (PE4), no name/EC/family; generic DEDDh exo, lacks IPR006309
PP_4768	Over-annotation → candidate_uncertain	DnaQ boilerplate but not ϵ ; lacks IPR006309; ExoX-like paralog
PP_3893	Over-annotation → candidate_uncertain	Phage-type DnaB (IPR019889); true helicase is dnaB/PP_4873
polA, ligA, rnhA, rnhB	Supporting (not core)	Primer processing / nick sealing; polA & ligA primary-bucketed to ppu03420
dnaC/dnaI, priB/priC/dnaT, seqA	not_expected_in_target_taxon	Enterobacterial-specific; genuine lineage divergence

The over-annotation calls are well-justified by the documented history of KT2440 re-annotation (Finding F006). The original genome ([PMID: 12534463](#)) was substantially revised: Belda et al. re-annotated 1548 gene functions and added 242 new protein-coding genes ([PMID: 26913973](#)), and GenBank-vs-RefSeq annotation discrepancies persist for this strain ([PMID: 41036861](#)). This establishes both the plausibility of over-propagated calls and the value of locus-level re-curation.

6. Module and GO-Curation Recommendations

Module verdict: COVERED (fully satisfiable). All seven steps have high-confidence gene assignments (Finding F008).

Step-by-step curation status:

- **Step 1 (initiation):** COVERED — add `dnaA` /PP_0010 to metadata.
- **Step 2 (unwinding + ssDNA):** COVERED — `dnaB` /PP_4873 + `ssb` /PP_0485.
- **Step 3 (priming):** COVERED — `dnaG` /PP_0388.
- **Step 4 (synthesis + proofreading):** COVERED — `dnaEA` /PP_1606 + `dnaQ` /PP_4141.
- **Step 5 (β -clamp):** COVERED — `dnaN` /PP_0011.
- **Step 6 (clamp loader):** COVERED — `dnaX` /PP_4269 + `holA` /PP_4796 + `holB` /PP_1966 + χ/ψ at PP_0978–PP_0979; `holE` / θ candidate_uncertain (not required).
- **Step 7 (ligation):** COVERED — `ligA` /PP_4274 (supported by `polA`, `rnhA`, `rnhB`).

Module-boundary guidance. The generic module boundaries are essentially correct for this organism, with two refinements: (i) the module should explicitly include the initiator DnaA even though KEGG map03030 omits it; and (ii) the primer-processing/nick-sealing enzymes (`polA`, `ligA`, `rnhA`, `rnhB`) should be labeled as *supporting* functions shared with repair (ppu03420), not core replisome subunits, to avoid double-counting across modules.

GO-curation notes. The broad EC 2.7.7.7 (DNA-directed DNA polymerase) mapping is attached to many clamp-loader subunits (`holA`/`holB`/`holC`) and to the exonuclease paralogs, which is misleading — these subunits are structural/ATPase components, not catalytic polymerases. Curators should prefer specific GO terms: DNA polymerase III complex (GO:0009360), DNA clamp loader activity, single-stranded DNA binding (`ssb`), 3'-5' exonuclease activity restricted to `dnaQ`, and DNA-directed DNA polymerase activity restricted to `dnaEA`. No new module documents or new GO term requests appear necessary; existing terms cover all satisfied steps.

7. Genes to Promote to Full Review

The following loci warrant full `fetch-gene` review:

1. **PP_0353 (Q88QY1)** — resolve identity of predicted "Exonuclease"; confirm ExoX/oligoribonuclease assignment and remove from replication module.
2. **PP_4768 (Q88DQ5)** — confirm DnaQ-family paralog (not epsilon); reassign away from Pol III core.
3. **PP_3893 (Q88G33)** — confirm phage/prophage-type DnaB paralog (IPR019889); reassign away from chromosomal replisome.
4. **PP_0978 / PP_0979 (Q88P75 / Q88P74)** — harmonize the χ (`holC`) / ψ (`holD`) annotation swap and assign gene names and protein names consistently.

5. **PP_0010 (P0A116, dnaA)** — add to module metadata as the Step-1 initiator (administrative, not a deep-review need, but must be entered).

8. Mechanistic Model / Synthesis

The KT2440 chromosomal replisome follows the canonical Gram-negative bacterial architecture with minimal lineage-specific modification within the core:

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Step 1: ORIGIN INITIATION
DnaA (PP_0010) -- binds oriC, opens duplex
[regulated by Hda/PP_1668, RIDA]
    |
    v
Step 2: FORK UNWINDING + ssDNA PROTECTION
DnaB helicase (PP_4873) --> 5'->3' unwinding
SSB (PP_0485) --> coats exposed ssDNA
[DnaC-independent loading -- no dnaC in KT2440]
    |
    v
Step 3: RNA PRIMING
DnaG primase (PP_0388) --> RNA primers
    |
    v
Step 4-6: POL III HOLOENZYME
+-----+
| a DnaEA (PP_1606) -- synthesis      |
| e DnaQ (PP_4141) -- 3'->5' proofread |
| b DnaN (PP_0011) -- processivity clamp loader: t/g DnaX (PP_4269) |
| d HolA (PP_4796), d' HolB (PP_1966) |
| x/psi PP_0978-PP_0979              |
+-----+
    |
    v
Step 7: OKAZAKI MATURATION
RNase HI/HII (PP_4142/PP_1605) -- primer removal
Pol I (PP_0123) -- gap filling
LigA (PP_4274) -- NAD+-dependent nick sealing

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NOT in core (over-annotations -> candidate_uncertain):

- PP_0353 predicted exonuclease (ExoX-like)
- PP_4768 DnaQ-family paralog (NOT epsilon)
- PP_3893 phage-type DnaB paralog (NOT replicative helicase)

Not expected in taxon: dnaC/dnaI, priB/priC/dnaT, seqA, holE

The discriminating diagnostic that resolves the three over-annotations is the **presence/absence of family-specific InterPro signatures superimposed on a shared fold**: the DEDDh exonuclease fold is shared by both the authentic proofreader and its paralogs, but only the true `dnaQ`/PP_4141 carries IPR006309; likewise the DnaB helicase fold is shared, but only the phage paralog PP_3893 carries IPR019889. This "shared-fold-plus-diagnostic-signature" logic is a reusable rule for future replication-module curation in other *Pseudomonas* strains.

9. Evidence Base

PMID	Title (abbreviated)	Relevance
12534463	<i>Complete genome sequence... P. putida KT2440</i>	Authoritative KT2440 genome underlying all PP_XXXX locus tags
26913973	<i>The revisited genome of P. putida KT2440</i>	1548 genes re-annotated, 242 new CDS added — justifies locus-level re-curation
41036861	<i>Differences in GenBank and RefSeq annotations</i>	Documents persistent annotation discrepancies in KT2440
30352930	<i>Segregation but not replication of the... chromosome</i>	Origin-proximal replication/segregation coordination context

Verified supporting quotations:

- **PMID: 26913973:** "We identified 242 new protein-coding genes and re-annotated the functions of 1548 genes, which are linked to almost 4900 PubMed references." — Direct evidence that KT2440 annotations have been substantially revised, justifying locus-level re-curation of the replication candidate genes.
- **PMID: 12534463:** "Sequence analysis of the 6.18 Mb genome of strain KT2440 reveals diverse transport and metabolic systems." — Establishes the authoritative KT2440 genome underlying the PP_XXXX locus tags used throughout this review.

Evidence-type summary. Every gene assignment in this review is grounded in **UniProt/InterPro/Pfam annotation plus genomic synteny**, cross-checked against the KEGG ppu03030 reference pathway and the re-annotated KT2440 genome. No direct target-strain biochemical or genetic experiments were located for the individual replisome subunits; transfer of mechanism from generic bacterial/*E. coli* models is **strong for the conserved core** (β -clamp, Pol III, DnaB, DnaG, SSB, LigA are near-universal) and **uncertain for lineage-specific accessories** (χ/ψ locus identity, holE presence, DnaC-independent loading).

10. Limitations and Knowledge Gaps

1. **Homology-based, not experimental.** Every gene assignment rests on annotation and synteny, not on direct biochemical or genetic experiments in KT2440. No target-strain

knockout, complementation, or in-vitro reconstitution data were found for the individual replisome subunits.

2. **hold ψ and hole θ are name-inferred.** ψ is inferred present from the PP_0978–PP_0979 synteny and size range, not from a confirmed gene name. θ (hole) may be genuinely absent; its dispensability makes this hard to distinguish from a gap.
3. **The χ/ψ annotation swap** was diagnosed from a UniProt inconsistency and has not been resolved against a curated reference — it needs a full gene-level review.
4. **Over-annotation calls** rest on signature analysis; final confirmation of the alternative functions (ExoX vs oligoribonuclease for PP_0353/PP_4768; prophage origin for PP_3893) would benefit from genomic-context inspection and phylogenetic placement.
5. **Literature specificity.** Most retrieved KT2440 literature concerns metabolic engineering, biodegradation, and genome annotation rather than replication mechanism directly; mechanistic conclusions for the conserved core are therefore "strong but indirect."

11. Proposed Follow-up Experiments / Actions

1. **Metadata fix (immediate):** Add `dnaA` /PP_0010 to the ppu03030 module as a covered Step-1 gene.
 2. **Full `fetch-gene` reviews** for PP_0353, PP_4768, PP_3893, and the PP_0978/PP_0979 pair (Section 7).
 3. **Genomic-context check** for PP_3893 — determine whether it lies within an annotated prophage region, which would corroborate the phage-DnaB call.
 4. **Phylogenetic placement** of PP_0353/PP_4768 among DEDDh exonucleases to assign ExoX vs oligoribonuclease vs other paralog identity.
 5. **GO-term refinement:** replace broad EC 2.7.7.7 mappings on clamp-loader subunits with specific complex/activity terms; restrict 3'-5' exonuclease activity to `dnaQ` /PP_4141.
 6. **Expert question:** confirm the *Pseudomonas* DnaC-independent helicase-loading mechanism so that the absence of `dnaC` is documented as expected rather than a gap.
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Review completed over 5 iterations; 8 findings confirmed; 12 papers reviewed. Module verdict: ppu03030 DNA replication is COVERED (fully satisfiable) in P. putida KT2440.

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